

TUGUEGARAO, Philippines

WMO: 982330

Lat: 17.6478N

Lon: 121.7585E

Elev: 70

StdP: 14.66

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	65.9	67.4	61.5	81.7	67.7	63.1	86.6	68.8	15.5	79.6	13.2	76.1	2.1	0	0.272

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	17.1	98.5	82.2	96.9	81.8	95.2	81.4	83.7	95.7	82.8	94.4	82.1	93.3	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.4	158.4	92.1	79.6	154.1	90.8	79.0	150.8	89.8	47.8	96.0	46.8	94.9	46.1	93.8	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.9	11.2	9.0	62.6	100.9	2.8	1.7	60.7	102.1	59.0	103.2	57.5	104.1	55.5	105.4		
			60.2	85.4	2.4	1.8	58.5	86.7	57.0	87.7	55.7	88.7	53.9	90.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	75.5	77.4	80.7	84.8	85.9	86.0	84.0	83.7	83.1	80.9	79.2	76.3
	DBStd	4.56	3.35	3.62	3.53	3.24	2.89	2.13	2.47	2.40	2.26	2.62	2.83	2.90
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11489	792	768	950	1043	1113	1081	1054	1044	994	959	875	816
	CDD65	6014	327	348	485	593	648	631	589	579	544	494	425	351
	CDH74	58875	1777	2491	4411	6821	7632	7719	6553	6339	5700	4371	3224	1837
	CDH80	24029	392	843	1850	3327	3765	3791	2763	2529	2147	1387	883	352
Wind	WSAvg	2.6	2.5	2.4	2.8	2.7	2.8	2.7	2.6	2.5	2.5	3.0	2.4	2.5
Precipitation	PrecAvg	68.0	1.5	1.1	1.1	1.8	5.0	6.2	8.4	9.1	8.1	10.7	10.2	4.7
	PrecMax	106.1	8.1	7.2	5.3	9.4	16.1	16.4	18.7	20.9	18.5	25.9	37.8	17.8
	PrecMin	45.4	0.2	0.0	0.0	0.0	0.3	1.1	1.9	2.2	1.7	0.6	0.5	0.0
	PrecStd	18.4	1.7	1.5	1.2	2.1	3.1	3.1	4.2	5.2	4.1	6.7	9.3	4.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.3	92.6	95.9	100.2	100.9	99.6	97.5	95.6	94.2	93.3	91.2	88.7
		MCWB	79.1	80.3	81.7	81.5	82.1	83.7	82.9	82.0	81.7	81.0	80.5	78.8
	2%	DB	86.4	90.3	93.9	97.9	98.8	97.3	95.2	93.6	92.7	90.9	89.3	86.0
		MCWB	77.7	79.1	80.2	81.1	82.2	82.8	82.1	81.5	81.0	80.2	79.8	78.0
	5%	DB	84.2	88.1	91.9	95.9	96.8	95.6	93.3	92.1	91.4	89.1	87.3	83.9
		MCWB	76.7	77.9	79.5	80.8	81.5	82.2	81.4	81.0	80.5	79.6	79.0	77.1
	10%	DB	81.9	85.5	89.6	93.8	94.5	93.8	91.2	90.3	89.7	87.1	85.2	81.7
		MCWB	75.6	77.0	78.4	80.3	81.0	81.5	80.7	80.5	80.1	78.9	78.0	76.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.7	81.0	82.8	84.4	84.7	85.5	83.7	83.0	82.7	81.9	81.2	79.7
		MCDB	87.4	90.7	94.3	97.8	97.6	97.5	95.2	92.5	92.3	91.2	89.3	87.0
	2%	WB	78.2	79.9	81.3	82.8	83.4	83.6	82.6	82.0	81.7	80.8	80.1	78.4
		MCDB	85.4	89.1	92.5	95.6	96.4	95.4	93.6	91.6	91.0	89.3	87.7	84.7
	5%	WB	77.0	78.7	80.1	81.7	82.3	82.6	81.7	81.3	80.9	80.0	79.3	77.4
		MCDB	83.5	87.2	90.7	93.9	94.9	94.2	92.0	90.6	89.8	87.7	86.2	82.8
	10%	WB	75.8	77.3	78.9	80.6	81.2	81.7	80.8	80.6	80.2	79.1	78.3	76.4
		MCDB	81.4	84.7	88.4	92.0	92.6	92.5	90.3	89.5	88.3	86.0	84.4	81.0
Mean Daily Temperature Range	5% DB	MDBR	12.3	14.6	16.6	18.4	18.4	17.1	15.0	13.7	13.7	12.6	11.1	10.1
		MCDBR	15.7	19.0	20.3	21.1	21.3	19.4	18.1	16.3	16.3	15.8	15.2	13.7
		MCWBR	9.9	10.9	10.4	9.1	8.8	8.7	8.8	7.7	7.9	8.1	8.5	8.3
	5% WB	MCDBR	15.0	18.0	19.7	20.0	20.2	18.6	17.0	15.3	15.4	14.6	14.0	12.5
		MCWBR	10.0	11.4	11.4	10.5	10.0	9.0	8.9	7.8	8.0	8.2	8.5	8.0
Clear-Sky Solar Irradiance	taub	0.423	0.421	0.469	0.478	0.458	0.435	0.427	0.432	0.444	0.452	0.449	0.458	
	taud	2.361	2.374	2.236	2.231	2.318	2.390	2.413	2.391	2.358	2.318	2.319	2.257	
	Ebn at Noon	272	280	269	266	268	271	274	275	272	267	262	257	
	Edh at Noon	37	39	46	46	42	38	37	39	40	40	39	40	
All-Sky Solar Radiation	RadAvg	1107	1344	1570	1876	1866	1874	1696	1584	1568	1324	1122	940	
	RadStd	101	189	152	139	142	143	148	113	108	101	155	111	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	-0.59	-0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page